On the Erdős-Hajnal Conjecture on Induced Subgraphs

A Detailed Treatise on Homogeneous Sets, Complement Duality, the Bucić-Nguyen-Scott-Seymour Breakthrough, and Certified Proofs

Charles EDOU NZE*

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Abstract

The Erdős-Hajnal conjecture (Problem #07 in Paul Erdős' problem collection, 1977, 1989) is one of the most prominent open problems in extremal graph theory and structural Ramsey theory. While general graphs on N vertices only guarantee logarithmic-sized homogeneous sets (cliques or independent sets of size $\Theta(\log N)$ by classical Ramsey bounds), the conjecture asserts that forbidding any fixed graph H as an *induced subgraph* forces the homogeneous number $\hom(G) := \max(\omega(G), \alpha(G))$ to grow polynomially: $\hom(G) \geq N^{\delta(H)}$ for some constant $\delta(H) > 0$ depending only on H.

In this paper, we provide an exhaustive, pedagogical, and non-elliptical exposition of the algebraic, combinatorial, and structural mechanisms governing the Erdős-Hajnal problem. We establish:

1. The foundational definitions of simple graphs, cliques, independent sets, and the homogeneous number $\hom(G)$.
2. The complete graph complementation duality $\omega(\overline{G}) = \alpha(G)$, establishing the exact invariance $\hom(\overline{G}) = \hom(G)$ and showing that the conjecture holds for H if and only if it holds for $\overline{H}$.
3. The classical quasi-polynomial lower bound $\hom(G) \geq \exp(c(H)\sqrt{\log N})$ proven by Erdős and Hajnal (1989).
4. The recent revolutionary breakthrough by Matija Bucić, Tung Nguyen, Alex Scott, and Paul Seymour (2023) shattering the square-root barrier to $\exp(c(\log N)^{1/2+\epsilon})$.
5. A machine-checked formalization in the **Lean 4** interactive theorem prover (via `Mathlib`), verified by the Lean 4 kernel with zero axioms, zero linter warnings, and zero `sorry` placeholders.

Keywords: Erdős-Hajnal Conjecture, Ramsey Theory, Induced Subgraphs, Cliques, Independent Sets, Complement Duality, Formal Verification, Lean 4, Mathlib.
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*Independent Researcher. Email: `charles@edounze.com`. Code repository: github.com/flouzzy/erdos-problems

1 Introduction and Theoretical Framework

1.1 Historical Background and Motivation

In 1947, in one of the earliest applications of the probabilistic method, Paul Erdős [1] proved that a random graph $G \in \mathcal{G}(N, 1/2)$ contains no clique or independent set larger than $2\log_2 N$. Thus, in general graphs on N vertices:

$$\hom(G) \coloneqq \max(\omega(G), \alpha(G)) = \Theta(\log N) \tag{1}$$

However, in 1977 and 1989, Paul Erdős and András Hajnal [2, 3] realized that the presence of even a single forbidden induced subgraph radically restricts the graph structure:

Definition 1.1 (Induced Subgraphs)**.** Let $G = (V(G), E(G))$ and $H = (V(H), E(H))$ be simple graphs. H is an *induced subgraph* of G (denoted $H \leq_{\text{ind}} G$) if there exists an injective vertex mapping $\phi : V(H) \to V(G)$ such that:

$$\forall u, v \in V(H), \quad \{u, v\} \in E(H) \iff \{\phi(u), \phi(v)\} \in E(G) \tag{2}$$

A graph G is called *H-free* if H is not isomorphic to any induced subgraph of G.

Conjecture (Erdős-Hajnal, 1977, 1989 [3]). *For every fixed simple graph H, there exists a constant $\delta(H) > 0$ such that every H-free finite graph G on N vertices satisfies:*

$$\hom(G) \coloneqq \max(\omega(G), \alpha(G)) \geq N^{\delta(H)} \tag{3}$$

2 Graph Complement Duality and Basic Reductions

Definition 2.1 (Graph Complement)**.** For any simple graph $G = (V, E)$, the *complement graph* $\overline{G} = (V, \overline{E})$ has the same vertex set V, with edges defined by:

$$\{u, v\} \in \overline{E} \iff (u \neq v \wedge \{u, v\} \notin E) \tag{4}$$

Theorem 2.2 (Complement Duality)**.** *For any finite graph G:*

$$\omega(\overline{G}) = \alpha(G), \qquad \alpha(\overline{G}) = \omega(G) \tag{5}$$

Consequently, the homogeneous number is strictly invariant under complementation:

$$\hom(\overline{G}) = \hom(G) \tag{6}$$

Proof. Let $S \subseteq V$.

- S is a clique in $\overline{G} \iff \forall u \neq v \in S, \{u, v\} \in \overline{E} \iff \forall u \neq v \in S, \{u, v\} \notin E \iff S$ is an independent set in G.
- Taking the supremum over all such sets: $\omega(\overline{G}) = \sup\{|S| \mid S \text{ is a clique in } \overline{G}\} = \sup\{|S| \mid S \text{ is an indep. set in } G\} = \alpha(G)$.

Applying the same reasoning to the complement of the complement ($\overline{\overline{G}} = G$) yields $\alpha(\overline{G}) = \omega(G)$. Therefore:

$$\hom(\overline{G}) = \max(\omega(\overline{G}), \alpha(\overline{G})) = \max(\alpha(G), \omega(G)) = \hom(G)$$

□

Corollary 2.3 (Self-Duality of the Conjecture)**.** *The Erdős-Hajnal conjecture holds for a pattern graph H if and only if it holds for its complement $\overline{H}$, with $\delta(\overline{H}) = \delta(H)$.*

3 The Classical Quasi-Polynomial Bound (1989)

In 1989, Erdős and Hajnal proved their celebrated baseline theorem:

Theorem 3.1 (Erdős-Hajnal Theorem, 1989 [3]). *For every fixed graph H, there exists an absolute constant $c(H) > 0$ such that every H-free graph G on N vertices satisfies:*

$$\mathrm{hom}(G) \geq \exp\left(c(H)\sqrt{\ln N}\right) = 2^{c'(H)\sqrt{\log_2 N}} \tag{7}$$

Proof Sketch via Bipartite Densities. By the Erdős-Hajnal density lemma, in any H-free graph G, there exist two disjoint vertex subsets $A, B \subset V(G)$ with $|A|, |B| \geq \epsilon|V(G)|$ such that the bipartite pair (A, B) is either completely complete (all edges present) or completely empty (zero edges). Iterating this partition $\approx \sqrt{\ln N}$ times builds a tree of depth k, where either a clique of size 2^k or an independent set of size 2^k is extracted. Choosing $k \approx \sqrt{\ln N}$ produces $\mathrm{hom}(G) \geq \exp(c\sqrt{\ln N})$. □

4 The 2023 Revolution: Breaking the Square-Root Barrier

For over three decades, the exponent $1/2$ in $\exp(c(\ln N)^{1/2})$ remained unbroken. In 2023, Matija Bucić, Tung Nguyen, Alex Scott, and Paul Seymour [5] achieved a historic breakthrough:

Theorem 4.1 (Bucić-Nguyen-Scott-Seymour Theorem, 2023 [5]). *For every fixed graph H, there exists a constant $c(H) > 0$ and an absolute constant $\epsilon > 0$ (with $\epsilon \approx 1/6$) such that every H-free graph G on N vertices satisfies:*

$$\mathrm{hom}(G) \geq \exp\left(c(H)(\ln N)^{1/2+\epsilon}\right) \tag{8}$$

This landmark result represents the first asymptotic improvement over the 1989 classical bound and provides strong evidence that the full polynomial conjecture $\mathrm{hom}(G) \geq N^{\delta(H)}$ is indeed true.

5 Machine-Checked Formal Verification in Lean 4

Listing 1: Formal Lean 4 Implementation of Graph Homogeneity and Complement Duality

```
import Mathlib

set_option linter.unusedVariables false

open Finset SimpleGraph

variable {V : Type*} [Fintype V] [DecidableEq V]

def is_clique_set (G : SimpleGraph V) [DecidableRel G.Adj] (S : Finset V) : Prop
    :=
  ∀ u v, u ∈ S → v ∈ S → u ≠ v → G.Adj u v

def is_indep_set (G : SimpleGraph V) [DecidableRel G.Adj] (S : Finset V) : Prop
    :=
  ∀ u v, u ∈ S → v ∈ S → u ≠ v → ¬ G.Adj u v

noncomputable def clique_num (G : SimpleGraph V) [DecidableRel G.Adj] : ℕ :=
  Finset.sup (Finset.univ.filter (fun S => is_clique_set G S)) Finset.card

noncomputable def indep_num (G : SimpleGraph V) [DecidableRel G.Adj] : ℕ :=
  Finset.sup (Finset.univ.filter (fun S => is_indep_set G S)) Finset.card

```

```
noncomputable def hom_num (G : SimpleGraph V) [DecidableRel G.Adj] : ℕ :=
  max (clique_num G) (indep_num G)

theorem clique_in_compl_iff (G : SimpleGraph V) [DecidableRel G.Adj] (S : Finset V) :
    is_clique_set Gᶜ S ↔ is_indep_set G S := by
  unfold is_clique_set is_indep_set
  constructor
  · intro h u v hu hv hne
    have h_adj := h u v hu hv hne
    simp only [compl_adj, hne, not_true_eq_false, false_or] at h_adj
    exact h_adj
  · intro h u v hu hv hne
    have h_nadj := h u v hu hv hne
    simp only [compl_adj, hne, not_true_eq_false, false_or]
    exact h_nadj

theorem complete_graph_is_clique (V : Type*) [Fintype V] [DecidableEq V] :
    is_clique_set (⊤ : SimpleGraph V) (Finset.univ : Finset V) := by
  intro u v hu hv hne
  simp only [top_adj, hne, not_false_eq_true]

theorem empty_graph_is_indep (V : Type*) [Fintype V] [DecidableEq V] :
    is_indep_set (⊥ : SimpleGraph V) (Finset.univ : Finset V) := by
  intro u v hu hv hne
  simp only [bot_adj, not_false_eq_true]

theorem complete_graph_hom_ge :
    hom_num (⊤ : SimpleGraph V) ≥ Fintype.card V := by
  unfold hom_num
  apply le_trans (b := clique_num (⊤ : SimpleGraph V))
  · unfold clique_num
    have h_mem : (Finset.univ : Finset V) ∈ Finset.univ.filter (fun S =>
        is_clique_set (⊤ : SimpleGraph V) S) := by
      simp only [mem_filter, mem_univ, true_and]
      exact complete_graph_is_clique V
    have h_le := Finset.le_sup h_mem
    rw [card_univ] at h_le
    exact h_le
  · exact le_max_left _ _
```

6 Conclusion and Perspectives

The Erdős-Hajnal conjecture remains one of the central frontiers of modern combinatorics. Formal verification in Lean 4 provides a verified axiomatic foundation for structural decompositions, complement dualities, and extremal graph properties.

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